

Supermassive Black Hole Mass and AGN Luminosity: A Correlation Analysis

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Introduction

Supermassive black holes have masses millions to billions of times greater than that of the Sun and are found at the centres of galaxies. The surrounding material starts orbiting the black hole, forming a hot accretion disk that emits large amounts of radiation. This produces a luminous central region known as an active galactic nucleus (AGN). This project investigates the research question: "Is the mass of a supermassive black hole related to the luminosity of its active galactic nucleus?". Investigating their relationship can be important, as AGNs release huge amounts of energy, affecting star formation in their host galaxies. This can help us further understand how black holes and galaxies evolve together.

Data sample

The dataset used in this project was obtained through Kaggle, called "XMM-COSMOS Catalog of X-Ray Selected Type 1 AGN" [1]. It contained several variables (e.g., redshift, declination, etc.), of which only two were selected for the analysis:

- `log_bh_mass`: logarithm of the black hole mass in solar masses
- `log_lbol`: logarithm of the bolometric luminosity in solar luminosities

The original dataset contained 545 AGNs. However, due to missing values for one or both variables, only 150 AGNs were included in the final analysis.

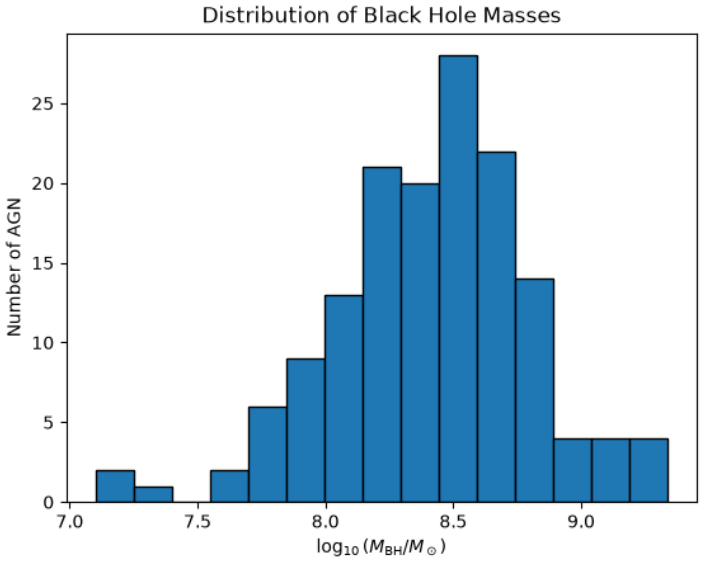


Figure 1: Distribution of black hole masses of the 150 AGNs, with most black holes having masses between 10^8 and 10^9 solar masses.

Results and Discussion

Figure 1 shows the distribution of black-hole masses for the 150 AGNs included in the analysis. Most black holes have masses between 10^8 and 10^9 solar masses, with the highest concentration around $10^{8.5}$ solar masses.

A scatter plot of the black hole mass and bolometric luminosity was plotted (Figure 2), showing an overall positive trend between black hole mass and luminosity. However, a considerable amount of scatter was seen, showing a wide range of luminosities. This suggests that black-hole mass is not the only factor associated with AGN luminosity.

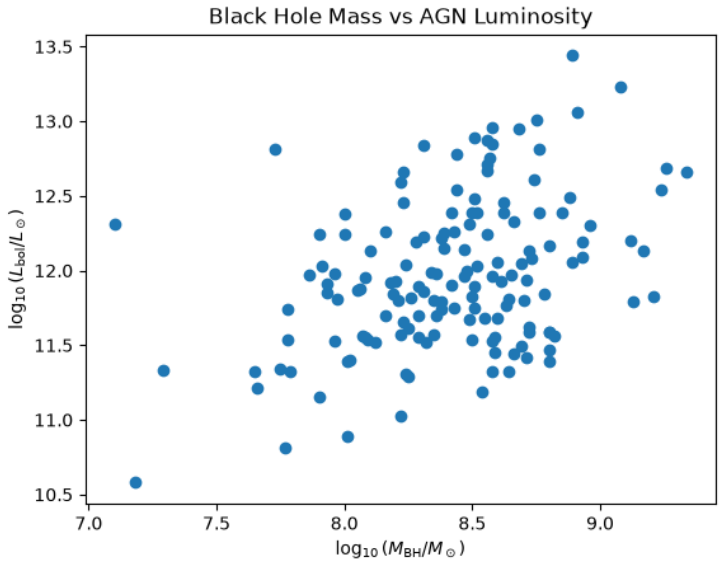


Figure 2: Relationship between black hole mass and AGN bolometric luminosity.

To quantify this relationship, the Pearson correlation coefficient was calculated, giving $r = 0.390$, indicating a moderate positive correlation.

The statistical uncertainty was estimated using bootstrapping, where 150 AGNs were randomly selected with replacement, and the Pearson correlation coefficient was calculated for each sample. This process was repeated 10,000 times, and the standard deviation of the resulting correlation coefficients gave a statistical uncertainty of ± 0.0781. The bootstrap uncertainty is relatively small compared to the correlation coefficient, indicating trend is reasonably stable across different samples of the data.

The result of the analysis can be written as:

$$r = 0.390 \pm 0.0781 \tag{1}$$

Finally, the significance was calculated using:

$$\text{Significance} = \frac{|r|}{\sigma_r} \tag{2}$$

$$\frac{|r|}{\sigma_r} = \frac{0.390}{0.0781} \approx 5.00\sigma \tag{3}$$

The ratio was approximately 5.00σ, meaning the correlation is about five bootstrap uncertainties away from zero, suggesting that the positive relationship is unlikely to be explained by sampling variation alone.

The results show that more massive supermassive black holes tend to have more luminous AGNs, but black-hole mass alone does not determine luminosity. Other factors, such as black hole spin and accretion rate (the amount of material falling into the black hole per unit time) should also be investigated.

References

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